

Two Vector Spaces That Are Not $\mathbb{R}^m$

Consider a vector space with the same scalars and vectors as in $\mathbb{R}^2$

$$\text{scalars: } \alpha \in \mathbb{R} \quad \text{and} \quad \text{vectors: } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad x_1 \in \mathbb{R}, \quad x_2 \in \mathbb{R}$$

but with scalar multiplication and vector addition defined differently

$$\alpha \mathbf{x} \equiv \begin{pmatrix} \alpha(x_1 - 1) + 1 \\ \alpha(x_2 - 1) + 1 \end{pmatrix} \quad \text{and} \quad \mathbf{x} + \mathbf{y} \equiv \begin{pmatrix} x_1 + y_1 - 1 \\ x_2 + y_2 - 1 \end{pmatrix}.$$

Let's refer to this *strange* vector space as S^2 . Of course, in order to call S^2 a vector space, it must be verified that structural conditions (a-0) through (d-2) listed in your previous homework are in fact true. Clearly, (a-0) is true. Moreover, vector addition is associative and commutative

$$(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \begin{pmatrix} x_1 + y_1 - 1 \\ x_2 + y_2 - 1 \end{pmatrix} + \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 - 1 + z_1 - 1 \\ x_2 + y_2 - 1 + z_2 - 1 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 + z_1 - 2 \\ x_2 + y_2 + z_2 - 2 \end{pmatrix},$$

$$\mathbf{x} + (\mathbf{y} + \mathbf{z}) = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 + z_1 - 1 \\ y_2 + z_2 - 1 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 + z_1 - 1 - 1 \\ x_2 + y_2 + z_2 - 1 - 1 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 + z_1 - 2 \\ x_2 + y_2 + z_2 - 2 \end{pmatrix},$$

$$\mathbf{y} + \mathbf{x} = \begin{pmatrix} y_1 + x_1 - 1 \\ y_2 + x_2 - 1 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 - 1 \\ x_2 + y_2 - 1 \end{pmatrix} = \mathbf{x} + \mathbf{y}.$$

You'll be asked to verify (a-3) and (a-4) in an exercise below. FYI:

$$\text{the additive identity } \mathbf{0} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{x}'\text{'s additive inverse } \mathbf{x}' = \begin{pmatrix} 2 - x_1 \\ 2 - x_2 \end{pmatrix}.$$

Continuing with the listed items, (m-0) is clearly true. For (m-1)

$$\begin{aligned} \alpha(\beta \mathbf{x}) &= \alpha \begin{pmatrix} \beta(x_1 - 1) + 1 \\ \beta(x_2 - 1) + 1 \end{pmatrix} = \begin{pmatrix} \alpha(\beta(x_1 - 1) + 1 - 1) + 1 \\ \alpha(\beta(x_2 - 1) + 1 - 1) + 1 \end{pmatrix} \\ &= \begin{pmatrix} \alpha\beta(x_1 - 1) + 1 \\ \alpha\beta(x_2 - 1) + 1 \end{pmatrix} = (\alpha\beta)\mathbf{x}, \end{aligned}$$

and (m-2)

$$1\mathbf{x} = \begin{pmatrix} 1(x_1 - 1) + 1 \\ 1(x_2 - 1) + 1 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \mathbf{x}.$$

For (d-1)

$$\alpha(\mathbf{x} + \mathbf{y}) = \alpha \begin{pmatrix} x_1 + y_1 - 1 \\ x_2 + y_2 - 1 \end{pmatrix} = \begin{pmatrix} \alpha(x_1 + y_1 - 1 - 1) + 1 \\ \alpha(x_2 + y_2 - 1 - 1) + 1 \end{pmatrix},$$

$$\begin{aligned} \alpha\mathbf{x} + \alpha\mathbf{y} &= \begin{pmatrix} \alpha(x_1 - 1) + 1 \\ \alpha(x_2 - 1) + 1 \end{pmatrix} + \begin{pmatrix} \alpha(y_1 - 1) + 1 \\ \alpha(y_2 - 1) + 1 \end{pmatrix} = \begin{pmatrix} \alpha(x_1 - 1) + 1 + \alpha(y_1 - 1) + 1 - 1 \\ \alpha(x_2 - 1) + 1 + \alpha(y_2 - 1) + 1 - 1 \end{pmatrix} \\ &= \begin{pmatrix} \alpha(x_1 + y_1 - 1 - 1) + 1 \\ \alpha(x_2 + y_2 - 1 - 1) + 1 \end{pmatrix}, \end{aligned}$$

and so $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$. You'll be asked to verify property (d-2) in an exercise below.

1. Let $\mathcal{V}$ be any vector space.

(a) Suppose $\mathbf{0}$ and $\mathbf{0}'$ are additive identities for $\mathcal{V}$. Show $\mathbf{0}' = \mathbf{0}$ to conclude $\mathcal{V}$'s additive identity is unique.

(b) Let 0 denote $\mathcal{V}$'s scalar field additive identity and let $\mathbf{0}$ denote $\mathcal{V}$'s vector additive identity. For any $\mathbf{x} \in \mathcal{V}$ show that $0\mathbf{x} = \mathbf{0}$.

(c) Suppose $\mathbf{x} \in \mathcal{V}$ has additive inverses $\mathbf{x}'$ and $\mathbf{x}''$. Show $\mathbf{x}'' = \mathbf{x}'$ to conclude $\mathbf{x}$'s additive inverse is unique.

(d) Let -1 denote $\mathcal{V}$'s scalar field additive inverse of its multiplicative identity 1 . For any $\mathbf{x} \in \mathcal{V}$ show that $-1\mathbf{x} = \mathbf{x}'$ where $\mathbf{x}'$ is $\mathbf{x}$'s additive inverse.

Please justify each step by stating which properties from (a-0) through (d-2) were used.

2. Recall what I called S^2 above.

(a) Explicitly calculate the additive identity vector, $\mathbf{0}$, and the additive inverse of $\mathbf{x}$, $\mathbf{x}'$, in order to establish that S^2 satisfies vector space properties (a-3) and (a-4).

(b) Show S^2 satisfies property (d-2).

(c) For S^2 confirm by calculating that $0\mathbf{x} = \mathbf{0}$ and $-1\mathbf{x} = \mathbf{x}'$.

3. Consider the following three vectors in our strange vector space S^2 :

$$\mathbf{0} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

Recall from the previous exercise that $\mathbf{0}$ here is S^2 's additive identity.

(a) Show on S^2 that the two vectors $\{\mathbf{x}, \mathbf{y}\}$ forms an independent set. Hint: You must conclude $\alpha\mathbf{x} + \beta\mathbf{y} = \mathbf{0} \iff \alpha = \beta = 0$. Remember how scalar multiplication and vector addition are defined on S^2 .

(b) Show that $\text{span}\{\mathbf{x}, \mathbf{y}\} = S^2$. Hint: For any $\mathbf{z} \in S^2$ I computed that

$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \quad \alpha\mathbf{x} + \beta\mathbf{y} = \mathbf{z} \quad \text{where } \alpha = z_1 - 3z_2 + 2 \text{ and } \beta = z_2 - 1.$$

4. Now consider the following two vectors in S^2 :

$$\mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

(a) Show that $\{\mathbf{x}, \mathbf{y}\}$ forms a dependent set on S^2 . Hint: I used $2\mathbf{x} = \mathbf{y}$.

(b) Describe a one parameter family of vectors which gives $\text{span}\{\mathbf{x}, \mathbf{y}\}$. Hint: I used the fact that $\alpha\mathbf{x} + \beta\mathbf{y} = \alpha\mathbf{x} + \beta 2\mathbf{x} = (\alpha + 2\beta)\mathbf{x} = \tilde{\alpha}\mathbf{x}$.

Let $\mathcal{P}_n$ denote the set of real polynomials of degree less than or equal to n ; $n \geq 0$. Let scalars be given by real constants. Define scalar multiplication by pointwise multiplication of functions by constants

$$\alpha \in \mathbb{R}, \quad \mathbf{p} \in \mathcal{P}_n, \quad \alpha \mathbf{p} \equiv \alpha p(x),$$

and vector addition by pointwise addition of functions

$$\mathbf{p} \in \mathcal{P}_n \text{ and } \mathbf{q} \in \mathcal{P}_n, \quad \mathbf{p} + \mathbf{q} \equiv p(x) + q(x).$$

The vector additive identity is the zero function. Constant functions including the zero constant function are taken to be polynomials of degree zero. With this, it's a routine exercise to show $\mathcal{P}_n$ is a vector space.

Clearly, the set of polynomials $\{1, x, \dots, x^n\}$ is a spanning set of vectors for $\mathcal{P}_n$. That is

$$\mathbf{p} \in \mathcal{P}_n \quad \Rightarrow \quad \mathbf{p} = p(x) = \alpha_0 1 + \alpha_1 x + \dots + \alpha_n x^n$$

for certain scalars $\alpha_0, \alpha_1, \dots, \alpha_n$. You'll be asked in exercise 5 below to show

$$\alpha_0 1 + \alpha_1 x + \dots + \alpha_n x^n = 0 \text{ for all } x \quad \Longleftrightarrow \quad \alpha_0 = \alpha_1 = \dots = \alpha_n = 0,$$

which says $\{1, x, \dots, x^n\}$ is an independent set. Therefore, $\{1, x, \dots, x^n\}$ is a basis (and in some sense the standard basis) for $\mathcal{P}_n$. Moreover, the dimension of $\mathcal{P}_n$ is $\dim(\mathcal{P}_n) = n+1$.

Here's another basis for the space $\mathcal{P}_2$. I'm taking $n = 2$ in order to make the present example easier to read. Consider

$$B = \{(x-1)(x-2), (x-0)(x-2), (x-0)(x-1)\}.$$

FYI: This is the *Lagrange basis* for polynomial interpolation at $x = 0, 1, 2$. One easily sees $\text{span } B$ is a subspace of $\mathcal{P}_2$. Moreover, B is an independent set. To see this, assume

$$b(x) = \alpha_0 (x-1)(x-2) + \alpha_1 (x-0)(x-2) + \alpha_2 (x-0)(x-1) = \mathbf{0} = 0 \text{ for all } x.$$

But $b(x) = 0$ for all x implies in particular $b(0) = b(1) = b(2) = 0$. However, $b(0) = 0$ implies $2\alpha_0 = 0$, and $b(1) = 0$ implies $-\alpha_1 = 0$, and $b(2) = 0$ implies $2\alpha_2 = 0$. Therefore $\alpha_0 = \alpha_1 = \alpha_2 = 0$ which says B is independent. Let me again state (but prove later) the following general vector space result.

$$\text{If } \mathcal{S} \subseteq \mathcal{V} \text{ and } \dim(\mathcal{S}) = \dim(\mathcal{V}) \text{ then } \mathcal{S} = \mathcal{V}.$$

Returning to our example, since $\text{span } B \subseteq \mathcal{P}_2$ and $\dim(\text{span } B) = 3 = \dim(\mathcal{P}_2)$, we can use the stated result to conclude $\text{span } B = \mathcal{P}_2$. Therefore, B is a basis for $\mathcal{P}_2$.

Here's one more potential basis for $\mathcal{P}_2$, this one I just made up out of the blue.

$$B = \{x-1, x+1, x^2+2\}.$$

This set of polynomial vectors doesn't have the nice structure of the standard basis or Lagrange basis we worked with above. Regardless, let's check what

$$b(x) = \alpha_0(x-1) + \alpha_1(x+1) + \alpha_2(x^2+2) = 0 \quad \text{for all } x$$

implies about the scalars α_0 , α_1 and α_2 . I can rewrite $b(x)$ in terms of the standard basis

$$b(x) = (-\alpha_0 + \alpha_1 + 2\alpha_2)1 + (\alpha_0 + \alpha_1)x + \alpha_2x^2$$

and use the fact that $\{1, x, x^2\}$ is independent to conclude

$$b(x) = 0 \quad \text{for all } x \quad \iff \quad -\alpha_0 + \alpha_1 + 2\alpha_2 = 0, \quad \alpha_0 + \alpha_1 = 0, \quad \alpha_2 = 0.$$

Now, use your Gaussian elimination skills to see

$$\begin{aligned} -\alpha_0 + \alpha_1 + 2\alpha_2 &= 0 \\ \alpha_0 + \alpha_1 &= 0 &\implies & \alpha_2 = 0, \quad \alpha_1 = 0, \quad \alpha_0 = 0. \\ \alpha_2 &= 0 \end{aligned}$$

Therefore, $\{(x-1), (x+1), (x^2+2)\}$ is an independent set.

5. Show $\alpha_0 1 + \alpha_1 x + \cdots + \alpha_n x^n = 0$ for all x implies $\alpha_0 = \alpha_1 = \cdots = \alpha_n = 0$. This says $\{1, x, \dots, x^n\}$ is a independent set in $\mathcal{P}_n$. Hint: Differentiate a certain number of times and then set $x = 0$.

6. Let x_0, x_1, x_2 be three distinct real numbers.

(a) Show that

$$\{(x-x_1)(x-x_2), (x-x_0)(x-x_2), (x-x_0)(x-x_1)\}$$

forms an independent set in $\mathcal{P}_2$.

(b) Write an arbitrary $p(x) \in \mathcal{P}_2$ in terms of a linear combination of these three vectors.

Answer:

$$p(x) = p(x_0) \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} + p(x_1) \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + p(x_2) \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}.$$

7. Determine whether or not the given sets of vectors from $\mathcal{P}_2$ are independent.

$$(a) \{1, (x-0), (x-0)(x-1)\} \quad (b) \{2x+1, x^2, (x+1)^2\}$$

Answers: (a) Independent. (b) Dependent.

8. Recall the independent set of vectors from $\mathcal{P}_2$ I cooked up above:

$$\{x - 1, x + 1, x^2 + 2\}.$$

Write $p(x) = \alpha_0 (x - 1) + \alpha_1 (x + 1) + \alpha_2 (x^2 + 2)$ by solving for the constant scalars α_0 , α_1 and α_2 when

$$(a) \ p(x) = 1 \quad (b) \ p(x) = x \quad (c) \ p(x) = x^2 \quad (d) \ p(x) = x^2 + 3x + 4$$

Answers: (a) $\alpha_0 = -1/2$, $\alpha_1 = 1/2$, $\alpha_2 = 0$. (b) $\alpha_0 = 1/2$, $\alpha_1 = 1/2$, $\alpha_2 = 0$.
(c) $\alpha_0 = 1$, $\alpha_1 = -1$, $\alpha_2 = 1$.

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